Traffic Light classification rules

Epstein, The Food & Activity Reference Guide (2012), as implemented in the Traffic Light Food Logger.

Every threshold and serving size in these tables is editable in the app's Settings. Page numbers refer to the guide.

1. The colour lines, by food group

Calories in one Traffic Light serving decide the colour. "Serving used" is the app's gram estimate of the guide's household serving; it is what the calories are measured against.

Food group	FRG serving	Serving used (g)	Green (cal)	Yellow (cal)	Red (cal)
Vegetables	1/2 cup	65	**30 or less**	31 to 45	above 45
Starchy vegetables	1/2 cup	80	n/a	120 or less	above 120
Fruit	1 small piece or 1/2 cup	90	n/a	90 or less	above 90
Fruit juice & dried fruit	8 fl oz, or 1/4 cup dried	240	n/a	n/a	**always red**
Grains	1 oz (1 slice; 1/2 cup cooked)	28	n/a	120 or less	above 120
Cold cereal	1 cup flakes, 1/2 cup pieces or 1/4 cup granola	30	n/a	120 or less	above 120, or above 25% of calories from sugar
Cheese	1 oz	28	n/a	83 or less	above 83
Milk, yogurt & other dairy	1 cup	245	n/a	135 or less	above 135
Meat & seafood	3 oz	85	n/a	158 or less	above 158
Eggs, beans & meat substitutes	1/2 cup, 1 egg or 1 patty	90	n/a	120 or less	above 120
Nuts & seeds	1 oz	28	n/a	68 or less	above 68
Fats, oils, sweets & others	about 1 Tbsp, or 1 oz of a snack	15	n/a	n/a	**always red**
Soup: broth-based & chowder	1 cup	245	n/a	186 or less	above 186
Soup: chili, bean/lentil & gumbo	1 cup	245	n/a	224 or less	above 224
Condiments, dressings & ingredients	1 Tbsp	15	n/a	23 or less	above 23
Combination food	no serving given	n/a	n/a	judged by its parts	holds 1 full serving of a red food
Water, plain tea/coffee, seasonings	n/a	n/a	**no colour**	n/a	n/a

2. Rules that override the calorie count

These are applied whatever the calories say.

Rule	What it means	Guide
Only vegetables can be green	No other group has a green line at all. A food in any other group is yellow at best.	p.6
Fruit juice and dried fruit are always red	Whatever the calories. Vegetable juice is not: it is judged on the vegetable range at 8 fl oz.	p.16-17, p.24
The whole Fats, Oils, Sweets & Others group is always red	Includes diet soda and anything artificially sweetened (sucralose, aspartame, acesulfame, stevia and the rest), and zero-calorie flavoured or sports waters.	p.2, p.59-66
Water, plain tea and coffee, and seasonings carry no colour	Plain means nothing added. Anything sweetened is red.	p.7
Cold cereal has one nutrient rule	Red when more than 25% of its calories come from sugar, even if the calories are within the yellow line.	p.77
A combination food gets no calorie range	It is red if it holds one full serving of a red food, and that serving can be made up of one red food or several together.	p.8

3. How the app works out which group a food is in

The colour follows from the group, so this is where most of the work happens. The guide names the groups but does not say how to recognise a supermarket product as one of them; everything in this section is the app's own reading, and the clinician can change the group afterwards with one dropdown.

What it looks at

Input	What it is	Example
Category tags	The food record's category, converted to tags that look like breakfast-cereals or skimmed-milks. For a USDA record these are produced from its foodCategory and description.	“Breakfast Cereals” becomes breakfast-cereals, which reaches step 4 and brings the sugar rule with it
The product name	Split into whole words.	“Carrots, raw” gives carrots and raw, which matches the vegetable words at step 14
The USDA category string	Matched as words, separately from the tags.	“Vegetables and Vegetable Products” carries a food that the name alone would not place

How it matches

Rule	Why it is written that way
Whole words only, never substrings	“pea” must not find “peanuts”, and the tag unsweetened-beverages must not read as “sweetened”. This one rule prevents most wrong colours.
Plural tolerance: s, es, and y to ies	“carrot” finds “Carrots, raw”; “berry” finds “berries”.
Plural tolerance cuts both ways	It is what lets “carrot” find “Carrots, raw”, and also what made “green” match the leafy-vegetable word “greens”, so that grapes described as “red or green” were read as a salad leaf and came out GREEN. A category that already says fruit now outranks a vegetable word found in the name.
A phrase matches consecutive words	“in heavy syrup” matches only where those three words run together, not wherever “syrup” appears.
First match wins	The steps in the next table are checked in order and the search stops at the first hit, so the order is part of the rule.

Guards checked before the groups

These stop a stray word in a product name from moving a food into the wrong group. Each one was added because a real product had been classified wrongly.

Guard	What it protects against
Grain identity	A grain-shaped product keeps its group when a stray word from another group appears in the name: “cinnamon raisin bagel” is not dried fruit, “Cape Cod popcorn” is not cod, “basmati rice with bell peppers” is not a vegetable. Chips and crisps are excluded, since the guide files them as snacks.
A name that says “sweet” wins	Cookie, brownie, cake, pastry, cinnamon roll, pudding and the rest override the grain words later on, so “oatmeal cookies” is a sweet. Rice cakes, pancakes, crab cakes and corn cakes are excluded from that list.
Meat and fish identity	A record the database files as meat, fish or a meat substitute keeps that identity when the name carries a sauce, oil or drink word: “sardines in tomato sauce”, “tuna with olive oil”, “beer bratwurst”.
Beans stay protein	Plain and dried beans, which USDA calls mature seeds, stay in the eggs, beans and meat substitutes group (p.41). Peanuts are excluded from the bean rule, because the guide files them with nuts (p.54-55).
Canned fruit is fruit	“In juice” or “in heavy syrup” makes it fruit at 1/2 cup, not a juice and not a sweet. Heavy syrup then lands red on calories alone (p.23).
Pack liquid is not the food	“Tuna in vegetable oil” and “artichoke hearts in oil” are not fats: the oil is what it is packed in.
A fruit is not a vegetable	A vegetable word found only in the name loses to a record already known to be fruit. Databases spell out colour variants (“Grapes, red or green, raw”, “Apples, red delicious”), and without this a colour word in a fruit name reaches the vegetable rule, which is the only rule that can return GREEN.
Inverted and technical names	Food databases do not write names the way a package does. USDA inverts them (“Beans, snap, green” rather than green beans), calls dried legumes “mature seeds”, files rice and pasta under “Cereal Grains and Pasta”, and writes hot oatmeal as “Cereals, oats, cooked”. Each of those has to be recognised explicitly, or the food lands in the wrong group: a vegetable in protein, a bean in nuts, or a grain in cold cereal, where it also picks up the sugar rule and the wrong serving.
Frozen desserts are sweets	Frozen yogurt, sherbet, sorbet, gelato and ice pops go to fats, oils and sweets rather than dairy (p.62).

4. The order the app checks food groups

The first match wins, so the order matters: it is what keeps ketchup out of the vegetable group and canned baked beans out of combination foods.

Step	What is checked
1	Drinks the guide files under Fats, Oils, Sweets & Others: red, diet versions included (p.2, p.59-60)
2	No colour: water, plain tea and coffee, seasonings (p.7). Anything sweetened is red instead.
3	Vegetable juice, judged as a vegetable at 8 fl oz (p.16-17). Fruit juice and dried fruit are red (p.24).
4	Granola and cereal bars as grains at 1 bar (p.27-28); hot cereal as a grain (p.28); cold cereal, which carries the sugar rule (p.77)
5	Sauces the guide files under vegetables and condiments. The sauce has to be the product itself.
6	Soups at 1 cup. Broth and stock are ingredients, not soups (p.73, p.75).

Step	What is checked
7	Combination foods: no range, only the one-red-serving rule (p.8)
8	Fats, Oils, Sweets & Others: always red (p.59-66)
9	Cheese at 1 oz, then milk, yogurt and the rest of dairy (p.33-40)
10	Nuts and seeds at 1 oz; nut butters at 2 Tbsp (p.54-55)
11	Condiments, dressings and ingredients at 1 Tbsp (p.73-76)
12	Produce disambiguation: green beans before legumes, starchy vegetables before other vegetables (p.12-20)
13	Meat and seafood at 3 oz, then eggs, beans and meat substitutes (p.41-58)
14	Vegetables at 1/2 cup; raw leafy greens weigh about 35 g per 1/2 cup
15	Fruit, yellow at most (p.21-23)
16	Grains at 1 oz dry, 1/2 cup cooked (p.26-32)
Last	A drink that matched nothing above: no colour if it is calorie-free and unsweetened, otherwise Fats, Oils, Sweets & Others (p.59)
None	Group not detected. The food is shown as yellow and the clinician picks the group from a dropdown.

5. Serving sizes that differ from the group default

When a food's name matches one of these, the app measures it at this serving instead of the group's. Where a range is given the label's own serving is used if it falls inside the range.

Food	Serving used (g)	Why
Cottage cheese	113	1/2 cup, against the cheese line (p.33, p.37)
Ricotta	62	1/4 cup
Cream cheese	29	2 Tbsp
Yogurt, skyr	170	6 oz, or the cup as sold (80-300 g)
Kefir	245	1 cup
Sour cream	115	1/2 cup, the only serving that reproduces the guide's printed colours (p.35, p.38)
Peanut, almond, cashew, sunflower butter, tahini	32	2 Tbsp (p.54-55)
Egg	50	1 egg
Tofu, tempeh	85	3 oz
Patty, veggie burger, links, nuggets	85	1 patty, or the label serving (40-150 g)
Edamame	78	1/2 cup
Baked, refried and pork and beans, beans in sauce	127	Beans in sauce are denser than drained beans
Pickles, spears	35	1 spear
Salsa, hummus, dip, guacamole	30	2 Tbsp

Food	Serving used (g)	Why
Gravy	60	1/4 cup
Whipped cream, whipped topping	3	1 Tbsp of aerosol cream weighs about 3 g (p.73)
Cocoa powder, baking cocoa	5	1 Tbsp (p.73)
Dates	24	1 medjool or 3 deglet noor, the guide's yellow serving (p.21)
Broth, stock, bouillon	245	1 cup, judged as an ingredient (p.73, p.75)
Vegetable juice	240	8 fl oz, on the vegetable range (p.16-17)
Lemon, lime, clam juice	15	1 Tbsp, as cooking ingredients (p.73)
Tomato, pasta, pizza sauce	60	1/4 cup, in the vegetable group
Fried potatoes	85	3 oz, as a starchy vegetable
Canned coconut milk	15	1 Tbsp, as an ingredient
Raw leafy greens	35	1/2 cup of spinach weighs half what 1/2 cup of carrots does
Canned fruit, applesauce, fruit cups	125	1/2 cup; heavy syrup lands red (p.23)
Cooked rice, pasta, hot cereal	75	1/2 cup
Filled pasta: ravioli, tortellini, gnocchi, pierogi	75	Sold ready to boil, so the 1/2 cup applies whatever the per-100 g value (p.31)
Instant oatmeal	43	1 packet, or the label packet (25-70 g)
Cereal and granola bars	28	1 bar, or the label bar (20-80 g)
Hot dogs, deli meat, sausage	85	No override: the guide judges every lunchmeat and sausage at 3 oz (p.44, p.52-53)

6. Daily limits and counting

Rule	Value	Source
Red servings allowed per day	2	“Eat 2 or less servings of RED foods per day” (p.3)
Default weekly red budget per client	14	2 a day across 7 days; editable per client
Red servings that make a combination food red	1	One serving, made up of one red food or several together (p.8)
Calories are rounded before comparing	whole calories	The guide's lines are whole calories and labels round too, so 120.4 is 120 and yellow
Servings counted, not items	FRG servings	The guide's serving is often not the label's: a plain bagel is 1/4 bagel in the guide and a whole bagel on the package

7. What the app never decides on its own

The guide is followed where it speaks and the clinician is asked where it does not. Nothing here is a guess presented as a verdict.

Situation	What happens
A packaged mixed dish	USDA publishes no percentage against any ingredient, so the app estimates one for each from the order they are listed in, which is the only thing a US label guarantees. It then colours each part and adds up the red servings. The estimate is shown with its arithmetic and can be overridden, and the dish is refused outright unless at least 60% of the pack can be priced.
A combination food it cannot take apart	The verdict is a fourth state, “needs a choice”, with a dashed badge and a ◆ mark. The app asks the guide's own question, “Does one serving hold a full serving of a red food?” Until it is answered the dish counts as nothing: it is left out of the green, yellow and red tallies and out of the weekly red budget.
The food group cannot be detected	The food is shown as yellow with the reason “Food group not detected”, and the clinician picks the group. No colour is asserted from a guess.
The record carries no calorie data	The food is shown as yellow and the reason says it could not be placed in the group's range.
Any verdict at all	Every verdict lists the reasons behind it, the food group can be changed, and the colour can be overridden per item.

Prototype, for professional use alongside clinical judgment. Not medical advice and not a medical device.

The seventeen groups, drawn to one scale

